United International University (UIU)

Dept. of Computer Science and Engineering (CSE)

Mid Exam Year: **2025** Semester: **Summer**

Course Code: **CSE 2217** Title: **Data Structure and Algorithms II**

Marks: **30** Time: **1 Hour 30 minutes**

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions. All questions are of values indicated on the right-hand margin.

1. **(a)** Determine the exact time cost in both the best and worst cases for each line of the following code. Then represent the total time complexity using asymptotic notation. (n, m > 0). **[4]**

```c
int ComputeSum(int a[], int b[], int n, int m){
    int s, sum = 0;
    for(int i = n/2; i<n; i++){
        s = 0;
        for(int j = 0; j < m ; j++){
            if(a[i] > b[j]){
                s = a[i];
                break;
            }
        }
        sum += s;
    }
    return sum;
}
```

(b) Suppose a **Divide and Conquer** algorithm **divides** an array of size n into 3 equal parts of size n/3 in **constant** time. Then, recursively solves the 3 subproblems. Finally, combine the result of the subproblems in **constant** time. If the array size is less than or equal to 2, it can solve the problem in **constant** time. **[1+2]**

i) Write the recurrence relation for time complexity of the above algorithm.

ii) Solve the recurrence relation using any method.

2 **(a)** A monitoring system records different types of alerts. The observed frequencies of these alerts are shown below:

Symbol	D	P	Y	M	E
Frequency	53	39	72	84	65

The system logs a total of **10,000 alerts.** Answer the following:

i) Build the **Huffman tree** for these alerts and **determine** the binary codeword for **each symbol**. Then encode the sequence **"DEMPY"** using your Huffman codes. **[3]**

ii) If a fixed-length binary code were used instead of Huffman coding, compute the **percentage storage savings** achieved by Huffman coding. **[2]**

b) A patient with diabetes is very hungry and has a bowl with a maximum capacity of 1000 mL. In front of him, there is a variety of fruits, each with a certain volume and sugar content. The patient may choose fractional portions of fruits if necessary. His goal is to completely fill the bowl while minimizing the total sugar intake. **[2+1]**

The available fruits are listed below:

Fruit	**Available Volume (mL)**	**Sugar Content (grams)**
Apple	200	36
Orange	300	42
Papaya	250	20
Mango	150	45
Watermelon	400	30

i) Using the **greedy** approach, determine which fruits (or fractions thereof) the patient should select to **fill his 1000 mL bowl** while consuming the **least amount of sugar**. Show your steps.

ii) Calculate the **total sugar consumed** with the optimal selection.

3 **(a)** Explain optimal substructure property and overlapping subproblems property in your own words. Why is dynamic programming not used for the maximum subarray problem? **[3]**

(b) You have $7 to spend on groceries. Being a health-conscious person who is currently trying to eat healthy, you want to buy items that are rich in protein. **[5]**

Upon reaching the grocery store, you come across several items and inspect their price and protein content (in grams). The information for each item is as follows:

Item	Price ($)	Protein (g)
Lentils (500g)	2	18
Milk (1L)	4	8
Eggs (6pcs)	3	15
Chicken (250 gm)	5	20

You can buy only one of each item. Which items should you buy to maximize your total protein intake without exceeding $7? What is the maximum total protein (in grams) you can get with this budget?

Note: To solve this problem, you must use the bottom-up dynamic programming approach. Show all the necessary calculations and values leading to your final answer.

4 **(a)** A factory along with using electricity from the national power grid also has a set of solar panels to produce electricity locally. The factory records its hourly net power usage as an array. **[4]**

Positive values = surplus energy produced (e.g., from solar panels).
Negative values = deficit (more energy consumed than produced, taken from national grid).

The factory wants to find the longest profitable window (maximum energy surplus).

Usage array (per hour, kWh):
[-2, -1, 6, -3, 5, -2, 4, -5]

Now, find out the most profitable stretch and show the detailed simulation to find the result.

(b) Write a pseudocode to calculate the sum of all numbers that are multiple of '3' in the given array with divide and conquer approach. **[3]**

Input Array: [9, 3, 0, 8, 7, 15, 20]